

Compact Modeling of RRAM Devices and Its Applications in 1T1R and 1S1R Array Design

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Abstract—In this paper, we present a compact model for metal-oxide-based resistive random access memory (RRAM) devices with bipolar switching characteristics. The switching mechanism relies on the dynamics of conductive filament growth/dissolution in the oxide layer. Besides the dc and pulsed I - V characteristics, the model also captures the RRAM retention property and the temperature dynamics. The model parameters and the device variations are calibrated from the experimental data of IMEC HfO_x -based RRAM devices. The model has been implemented in Verilog-A, which can be easily adapted into the SPICE simulator for the circuit-level analysis. As case studies, we demonstrate the model's applications on the programming scheme design of the 1-transistor-1-resistor array, as well as the design space exploration of the 1-selector-1-resistor cross-point array toward megabit-level.

Index Terms—1-selector-1-resistor (1S1R), 1-transistor-1-resistor (1T1R), compact model, conductive filament (CF), cross-point array, resistive random access memory (RRAM), resistive switching, selector, variations.

I. INTRODUCTION

THE METAL-OXIDE-BASED resistive random access memory (RRAM) has been considered as a promising candidate for the next-generation nonvolatile memory application due to its low programming voltage (<3 V), fast switching speed (<10 ns), excellent scalability (<10 nm), and good compatibility with the CMOS fabrication [1], [2]. RRAM is usually a two-terminal device with a simple metal-insulator-metal structure. Operation of RRAM devices relies on the voltage-driven resistance switching of the metal-oxide between the low-resistance state (LRS) and the high-resistance state (HRS), which is generally attributed with the formation and rupture of the conductive filament (CF) that consists of oxygen vacancies [3], [4]. The transition from HRS to LRS is called the SET process, and its reverse is called the RESET process. For bipolar RRAM, SET is achieved by applying a positive SET voltage, and RESET is achieved by applying a negative RESET voltage. To model these switching behaviors, it is important to include the underlying physics of RRAM switching mechanism in the model and validate

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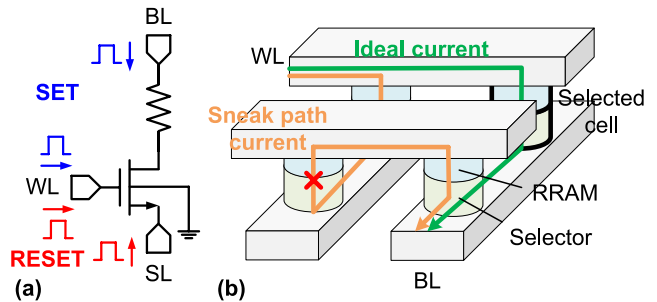


Fig. 1. (a) Circuit schematic of 1T1R configuration. (b) Illustration of 1S1R array structure. The selector can help cut off the sneak path current.

it with the experiments. Meanwhile, as the variations in the switching dynamics are generally critical factors that affect the application of RRAM devices, the model should also be able to predict the potential impact of variation, providing guidelines for the improvement of the device performance by minimizing these variations.

In general, there are two types of RRAM array architectures. The first one is the 1-transistor-1-resistor (1T1R) architecture, where the single-cell configuration is shown in Fig. 1(a). The top electrode (TE) of RRAM is connected to the bitline (BL). The gate of the transistor is controlled by the wordline (WL) with the source of the transistor connected to the source line (SL). The SET (RESET) operation is performed by applying voltage pulses at the WL and BL (SL), and the transistor bulk is grounded. The second one is the 1-selector-1-resistor (1S1R) cross-point array architecture, which was proposed as an alternative memory architecture to offer a smaller cell area ($4F^2$, F is the feature size) than the 1T1R architecture ($6F^2$ when $W/L = 1$). With the help of the selector, 8-Mb to 32-Gb cross-point array macros have been demonstrated [5], [6]. As shown in Fig. 1(b), the selector is typically a two-terminal device stacked in series of the RRAM device, which can be viewed together as one cell. In the cross-point architecture, the WLs and BLs are perpendicular to each other, and the cells are connected in between. The selector can effectively cut off the leakage current at low-voltage bias [7], which helps alleviate the well-known sneak path problem in the cross-point architecture [8], [9].

In circuit-level designs, compact models of RRAM devices are very useful for evaluating the feasibility of RRAM in various applications. Several existing RRAM compact models were developed to reproduce the detailed resistive switching characteristics [10]–[13], with the increased computational

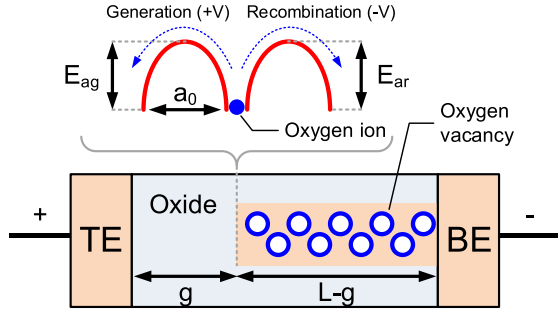


Fig. 2. Schematic of RRAM device structure and the generation/recombination process of oxygen ion at the CF tip.

complexity that might not be efficient for a circuit simulation. In this paper, we improve the Verilog-A RRAM compact model on top of our earlier works [14], [15]. The new result of the model includes the resistance retention failure mechanism that is essential for the RRAM reliability evaluation. We also implement the transient heat conduction process in the Verilog-A program, which is necessary for applications that require the ultrafast I - V responses, e.g., assessing the vulnerability of RRAM device under heavy ion irradiation [16]. The model parameters are validated with the experimental data of IMEC HfO_x -based RRAM devices [17]–[19]. The presented model in this paper is available for downloading at [20]. As case studies, we apply this model to two general circuit-level applications: one is the programming scheme design on the 1T1R configuration and the other is the 1S1R cross-point array design exploration for a large-scale array considering the device variations.

II. RRAM COMPACT MODEL

A. Modeling of CF Growth/Dissolution

The RRAM device structure in the compact model is shown in Fig. 2. The oxide layer is sandwiched between the TE and the bottom electrode. The switching characteristics are strongly associated with the distribution of oxygen vacancies in the oxide layer, where they act as the electron hopping sites for current conduction paths. In this model, we simplify the picture to be a single dominant CF in 1-D. The primary internal variable used in this model is the gap distance (g), which is defined as the average distance between the TE and the tip of the CF. g can determine the RRAM resistance through an electron tunneling conduction mechanism, where the resistance exponentially increases with g . Besides, the resistance also has the nonlinear relationship with the applied voltage (V). In general, the RRAM resistance shows linear dependence at small V (typically <0.5 V) and exponential dependence at large V , which can be modeled by a hypersinusoidal function. Thus, the I - V relationship of the RRAM model is expressed as

$$I = I_0 \exp\left(-\frac{g}{g_0}\right) \sinh\left(\frac{V}{V_0}\right) \quad (1)$$

where I_0 , g_0 , and V_0 are fitting parameters.

The RRAM switching is essentially a dynamic process, even with fixed V , and the current I is time-dependent as g evolves.

As shown in Fig. 2, by applying a positive (negative) voltage on the RRAM device, the CF will grow (dissolute) due to the generation (recombination) of oxygen vacancies and oxygen ions at the tip of the CF, which can be expressed with the following equations:

$$\frac{dg}{dt} = -v_0 \left[\exp\left(-\frac{qE_{ag}}{kT}\right) \exp\left(\frac{\gamma a_0 qV}{L kT}\right) - \exp\left(-\frac{qE_{ar}}{kT}\right) \exp\left(-\frac{\gamma a_0 qV}{L kT}\right) \right] \quad (2)$$

$$\gamma = \gamma_0 - \beta \left(\frac{g}{g_1}\right)^3 \quad (3)$$

$$g(t + dt) = g(t) + dg \quad (4)$$

where dg/dt is the gap growth/dissolution velocity. g has the lower and upper limit on the average gap size $g_{\min}$ and $g_{\max}$, respectively. $g_{\min}$ represents the tip of CF is nearly in contact with the TE during the SET. In this case, the resistance at the TE interface layer may become the dominant player. $g_{\max}$ stands for the residual CF, which cannot be removed anymore during the RESET. In (2), dg/dt can be calculated by the net difference between the oxygen vacancy generation and recombination rates. E_{ag} (E_{ar}) is the activation energy for oxygen ions to migrate from one potential well to another in the generation (recombination) process [4]. If E_{ag} is not equal to E_{ar} , the filament will still be gradually changing even under zero voltage bias, which accounts for the ion self-diffusion process. Normally, E_{ag} should be larger than E_{ar} [21], in order to capture the dominant retention failure mode (LRS resistance shift toward HRS) as reported in [18]. L is the oxide thickness, and a_0 is the atomic hopping distance. $a_0(qV/L)$ can be viewed as the energy barrier raising/lowering to the neighboring oxygen vacancy site. γ is the g -dependent local field enhancement factor, which is calculated in (3). It considers the polarizability of high- k dielectrics and the nonuniform potential distribution in the device structure [22]. The form of (3) is obtained from the empirical pulsed I - V fitting of HfO_x -based RRAM devices in [23], which introduces the positive (negative) feedback on the filament growth (dissolution) for abrupt SET and gradual RESET behaviors in typical bipolar RRAM devices. v_0 , γ_0 , β , and g_1 are fitting parameters. To link the RESET/SET operations (with applied field and temperature) and retention at various storage temperatures, temperature has to be introduced in the switching dynamics, as modeled by (2). For compact modeling in 1-D, we reduce the complex heat conduction process to a lumped thermal model with the effective thermal time constant (τ_{th}) and thermal capacitance (C_{th}). The temperature T is regarded as the local temperature around the filament tip, and its evolution can then be expressed as

$$\frac{dT}{dt} + \frac{T - T_0}{\tau_{\text{th}}} = \frac{|V \times I|}{C_{\text{th}}} \quad (5)$$

$$T(t + dt) = T(t) + dT \quad (6)$$

In (5), T_0 is the ambient temperature. In [14], the differential equations of (2) and (5) are implemented using a macrocircuit model. In this paper, the equation is formulated using the backward Euler method [24], thus, can be implemented in the

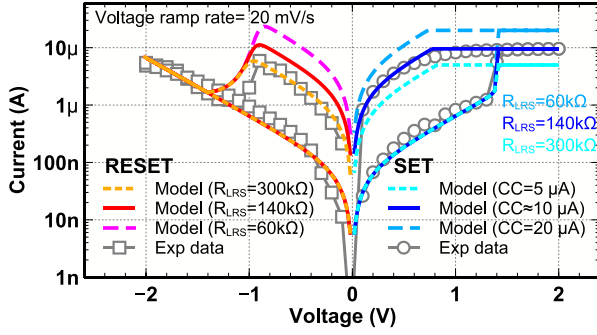


Fig. 3. DC I - V fitting with the IMEC HfO_x -based RRAM devices. The experimental data are fitted with the RESET process starting from $R_{\text{LRS}} = 300 \text{ k}\Omega$ followed by the SET process with $\text{CC} \approx 10 \mu\text{A}$.

TABLE I
PULSE PROGRAMMING FITTING IN 1T1R CONFIGURATION

	IMEC [17]	Model
SET ($R = 100 \text{ k}\Omega$ to $R = 10 \text{ k}\Omega$)	$V_{\text{WL}} = 1 \text{ V}$ $V_{\text{BL}} = 1.8 \text{ V}$ SET time $\leq 10 \text{ ns}$	$V_{\text{WL}} = 1.4 \text{ V}$ $V_{\text{BL}} = 1.8 \text{ V}$ SET time $\approx 10 \text{ ns}$
RESET ($R = 10 \text{ k}\Omega$ to $R = 100 \text{ k}\Omega$)	$V_{\text{WL}} = 3 \text{ V} \sim 3.5 \text{ V}$ $V_{\text{SL}} = 1.6 \text{ V} \sim 2.5 \text{ V}$ RESET time $\leq 10 \text{ ns}$	$V_{\text{WL}} = 3 \text{ V}$ $V_{\text{SL}} = 1.8 \text{ V}$ RESET time $\approx 7.5 \text{ ns}$

Verilog-A program to increase the computational efficiency and to reduce the complexity of the RRAM model.

The model parameters are calibrated from the IMEC $\text{TiN}/\text{Hf}/\text{HfO}_x/\text{TiN}$ -based RRAM devices [17]–[19]. The fitting parameters in (1) and the range of g are determined by the dc I - V experimental data shown in Fig. 3. To simulate the dc SET process in SPICE, a current limiter such as a transistor is added in series with the RRAM device to implement the compliance current (CC). The parameters in (2) and (3) are fitted to the experimental data in the pulsed programming condition of the 1T1R configuration [17] and the average LRS retention behavior in [18]. In this paper, the transistor is implemented using the Predictive Technology Model (PTM) model [25] at 130-nm technology, to match the channel length that was used in IMEC's test structures [17]. The biasing configuration used in the pulsed programming is shown in Table I, and the LRS retention, as shown in Fig. 4, with the resistance R is readout at $V = 0.1 \text{ V}$ (this definition of R applies to the rest of this paper). The thermal parameters in (5) are defined to ensure the RRAM device can reach its thermal equilibrium within a physically reasonable timescale ($\sim \text{ns}$) once it conducts current. The temperature range is typically from the room temperature to a few hundreds of degree celsius [26]. The parameter values used are listed in Table II.

B. Modeling of Variation in RRAM Devices

The variation in RRAM devices is a well-known problem as it will bring significant challenges into the circuit design with the RRAM integration. The variation generally comes from the stochastic nature of gap formation and evolution. Even an atomic fluctuation on the RRAM switching process will result in large differences in the resistance state. The variation

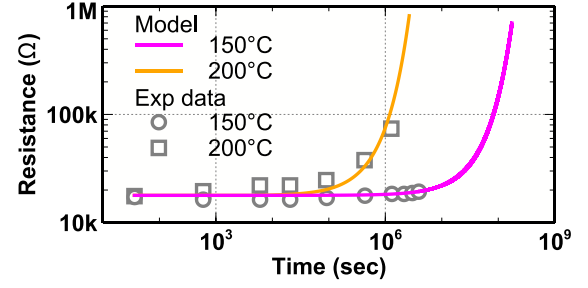


Fig. 4. Retention time fitting with the IMEC HfO_x -based RRAM devices using the experimental data from [18].

TABLE II
MODEL PARAMETERS

Symbol	Value	Symbol	Value
I_0	61.4 μA	g_0	0.275 nm
V_0	0.43 V	γ_0	16.5
β	1.25	g_1	1 nm
v_0	150 m/s	E_{ag}	1.501 eV
E_{ar}	1.5 eV	a_0	0.25 nm
L	5 nm	g_{min}	0.1 nm
g_{max}	1.7 nm	T_0	298 K
τ_{th}	0.23 ns	C_{th}	0.318 fJ/K

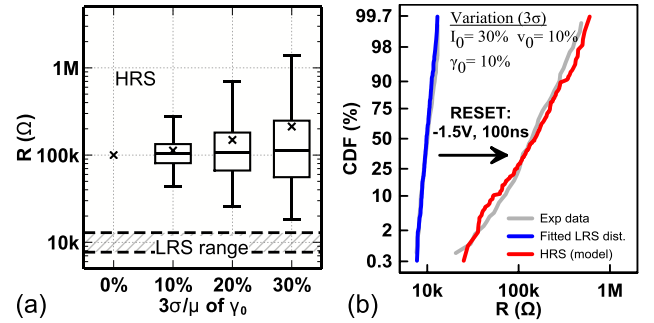


Fig. 5. (a) Impact of γ_0 variation on R_{HRS} after 10-ns RESET from $R_{\text{LRS}} = 10 \text{ k}\Omega$. The LRS distribution range is illustrated for reference. (b) Variation fitting of the IMEC HfO_x -based RRAM devices in the RESET process using the experimental data from [19].

is commonly seen in different devices, and the same device at different programming cycles [27]. Before introducing the variation into the model, it is worth investigating the sensitivity of the model parameters. As discussed before, g is the primary variable that determines the resistance with an exponential dependence, and dg/dt is strongly associated with the switching characteristics. Apparently, varying the parameters in either (1) or (5) only has limited effect in reflecting the large variability of RRAM devices. Instead, the parameters in (2) mostly reside in the exponential function, which are believed to have the greatest impact on the RRAM switching characteristics. Thus, γ_0 is selected to perform the sensitivity analysis to examine its impact on the HRS resistance (R_{HRS}). The devices are reset from a nominal LRS resistance $R_{\text{LRS}} = 10 \text{ k}\Omega$ to $R_{\text{HRS}} = 100 \text{ k}\Omega$. As shown in Fig. 5(a), with the same RESET voltage, the log-scale distribution of HRS becomes more and more diverse with increasing variation on γ_0 . The result is common in RRAM devices that the

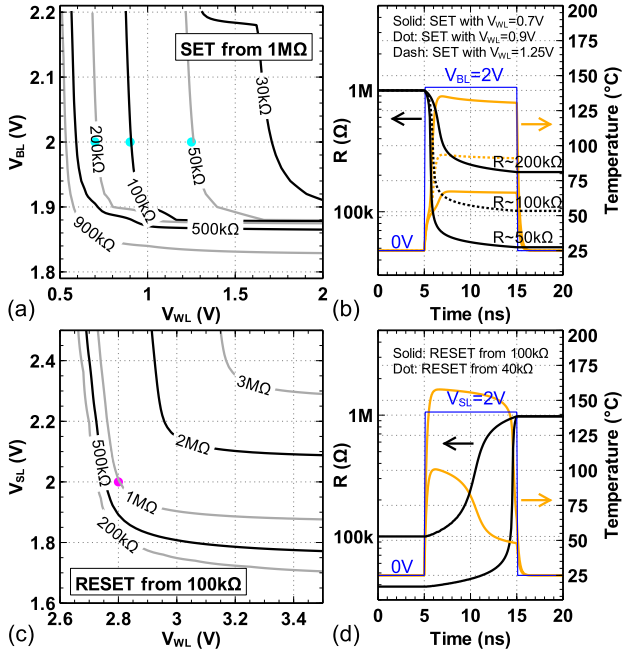


Fig. 6. (a) LRS contours as a function of V_{WL} and V_{BL} after 10-ns SET from $R_{HRS} = 1 \text{ M}\Omega$. (b) Multilevel states achieved by controlling V_{WL} in SET [light blue points in (a)] and the temperature evolutions. (c) HRS contours as a function of V_{WL} and V_{BL} after 10-ns RESET from $R_{LRS} = 100 \text{ k}\Omega$. (d) RESET from $R_{LRS} = 100 \text{ k}\Omega$ [purple point in (c)] and $R_{LRS} = 40 \text{ k}\Omega$ and the temperature evolutions.

HRS has a wider distribution, indicating varying the values of γ_0 alone may be able to approximately model the RRAM variability behaviors. Fig. 5(b) shows the variation fitting in the RESET using the experimental data from [19]. First, I_0 is chosen in (1) to fit the initial LRS distribution with an assumption that all the devices are initially at LRS where $g = g_{\min} = 0.1 \text{ nm}$. Then, γ_0 and v_0 in (2) and (3) are selected to fit the HRS distribution. The $3\sigma/\mu$ (3 standard deviation/mean) of I_0 , γ_0 , and v_0 are 30%, 10%, and 10%, respectively.

III. PROGRAMMING SCHEME DESIGN OF 1T1R CONFIGURATION

In this section, we demonstrate the application of the compact RRAM model with the 1T1R configuration to design the programming scheme for the SET and RESET operations. With the RRAM compact model, the 1T1R configuration can be simulated by SPICE to design the voltage pulse amplitudes for SET and RESET. In this paper, we assume the pulse widths are fixed (10 ns), and the variation of RRAM devices is not considered for simplicity. We use the minimal transistor W/L ratio = 1. Fig. 6(a) shows the LRS contours as a function of WL and BL voltage amplitude (V_{WL} and V_{BL}) after 10-ns SET from an initial HRS at $1 \text{ M}\Omega$. It can be seen that V_{WL} has much better control than V_{BL} , because varying values of V_{WL} can limit the amount of transistor current passing through the RRAM device. It acts as a current compliance to prevent the RRAM device being switched to the lowest LRS limit, leading to intermediate multilevel states in the SET process. As shown in Fig. 6(b), by selecting appropriate V_{WL} at 0.7, 0.9, and 1.25 V [Fig. 6(a) (light blue points)],

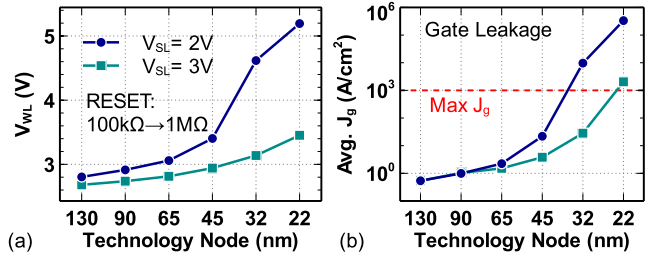


Fig. 7. (a) Required V_{WL} for different technology nodes of transistor to reset the device from $100 \text{ k}\Omega$ to $1 \text{ M}\Omega$. (b) Average gate current density using V_{WL} in (a). As the transistor size is scaled down, larger value of V_{WL} is required to increase the current drivability of the transistor. However, the induced large gate leakage will become an issue for the reliability of the transistor.

multilevel states at 200, 100, and $\sim 50 \text{ k}\Omega$ can be achieved with a fixed $V_{BL} = 2 \text{ V}$ pulse in 10 ns. The temperature evolutions for these multilevel states are also illustrated, where the peak temperatures are not beyond 150°C mainly due to the larger value of R_{LRS} used in this paper and the transistor current compliance effect. Fig. 6(c) shows the HRS contours as a function of V_{WL} and the SL voltage amplitude (V_{SL}) after 10-ns RESET from initial $R_{LRS} = 100 \text{ k}\Omega$. In this case, controlling V_{WL} and V_{BL} has similar effects to achieve multilevel states, and it seems that varying the V_{SL} value is effective in tuning the intermediate HRS, because RESET is a more gradual resistance switching process. Fig. 6(d) demonstrates the 10-ns RESET process and the temperature evolution from initial $R_{LRS} = 100 \text{ k}\Omega$ with V_{WL} at 2.8 V and V_{SL} at 2 V [Fig. 6(c) (purple points)] and $R_{LRS} = 40 \text{ k}\Omega$ with V_{WL} at 3.2 V and V_{SL} at 2 V. It should be noted that the SET state can last for more than 10^6 s without any external biasing at 150°C (Fig. 4), and can be switched to the HRS within 10 ns [Fig. 6(d)] when BL biasing is applied. This huge difference in the time scale is possible, mainly due to the bipolar RRAM's voltage-driven switching mechanism [4].

The other thing is that the RESET process normally requires larger values of V_{WL} than the SET in the 1T1R configuration, because in the RESET, part of V_{SL} drops across the RRAM, and thereby, the actual value of V_{GS} is smaller than the value of V_{WL} . As shown in Fig. 6(a) and (c), by fixing V_{BL} and V_{SL} at 2 V, the required V_{WL} for SET to $100 \text{ k}\Omega$ and RESET to $1 \text{ M}\Omega$ can have about 3 times difference. With the downscaling of the transistor size, the voltage bias in RESET may potentially cause transistor reliability issues, such as the gate dielectric breakdown and the p-n junction breakdown from the drain to bulk. The applied voltages actually need to be increased with the scaling as smaller sized transistors (with the same value of $W/L = 1$) needs larger value of V_{GS} to maintain similar current drivability. Fig. 7(a) shows the required RESET V_{WL} for different technology nodes of transistor using the PTM model. For the same RESET condition from $100 \text{ k}\Omega$ to $1 \text{ M}\Omega$, V_{WL} needs to be boosted to >5 and $\sim 3.5 \text{ V}$ for the transistor at 22 nm with $V_{SL} = 2$ and 3 V, respectively. Under these V_{WL} biases, the average gate leakage current density (J_g) is measured for both cases, as shown in Fig. 7(b). The result shows that J_g is increased to $>10^5$ and $>10^3 \text{ A/cm}^2$ at $V_{SL} = 2$ and 3 V, respectively. ITRS suggests that J_g of transistors should be less than 10^3 A/cm^2 even for high-performance applications [28], indicating the transistors

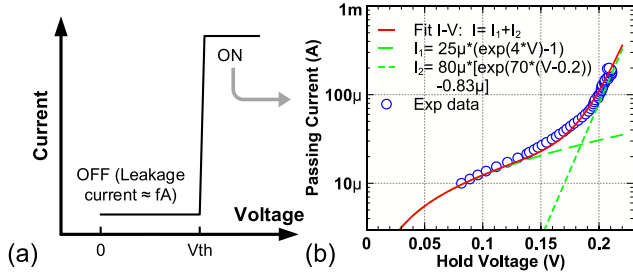


Fig. 8. (a) I - V characteristics of the FAST selector. V_{th} is the turn-ON threshold. (b) After switched ON, passing current as a function of hold voltage. The experimental data are extracted from [30] using the relation: hold voltage = passing current $\times$ resistance.

beyond ~ 32 nm may not meet the reliability requirement. Although V_{WL} can be reduced by boosting V_{SL} , a large V_{SL} value may also suffer from the reliability issue, such as the p-n junction breakdown. To alleviate the challenges of scaling the 1T1R to advanced technology nodes, specially designed transistor's gate-stack or further device engineering of RRAM characteristics is needed. For the RRAM optimization, the key efforts are needed to lower down the switching current from today's tens of microamperes to a few microamperes, which can relax the requirement of V_{WL} . Further lowering the RRAM switching voltages (V_{BL} and V_{SL}) down to 1 V is also helpful.

IV. 1S1R CROSS-POINT ARRAY DESIGN

In this section, we demonstrate the design exploration of 1S1R cross-point array for a large-scale array considering the variations in RRAM devices. In recent years, several selector devices have been proposed and demonstrated [29]–[32]. Among them, the field-assisted superlinear threshold (FAST) selector [30] exhibits excellent performance, which shows outstanding selectivity ($>10^{11}$), small switching slope (<5 mV/decade), and high current drivability (>5 MA/cm 2). More importantly, the threshold voltage (V_{th}) of FAST is claimed to be adjustable, with the reported range from ~ 0.3 to 1.3 V. Therefore, we extract the device characteristics from the FAST selector for modeling of the selector device in this 1S1R array design. The simplified I - V characteristics of the FAST selector is shown in Fig. 8(a). The threshold voltage (V_{th}) of the FAST selector is the turn-ON threshold, and the OFF-state leakage current of the selector is at $\sim$ fA level [30]. Once the selector switches ON, a small hold voltage is required to pass a specific current. As shown in Fig. 8(b), the relationship between the passing current and the hold voltage of the FAST selector can be fitted with the addition of two exponential functions. To avoid any convergence issues in SPICE simulators, we also constrain the fitting parameters to guarantee $I = 0$ at zero hold voltage in this curve.

Fig. 9 shows the bias scheme for the write and read operations of the 1S1R cross-point array. The array size is $N \times N$, and N represents the number of WLs and BLs. The $V/3$ bias scheme is used for both the write and read operations, because for the write operation, it can maintain the largest write margin, and the tremendous write leakage can be almost eliminated by the FAST selector [33]. For the read operation,

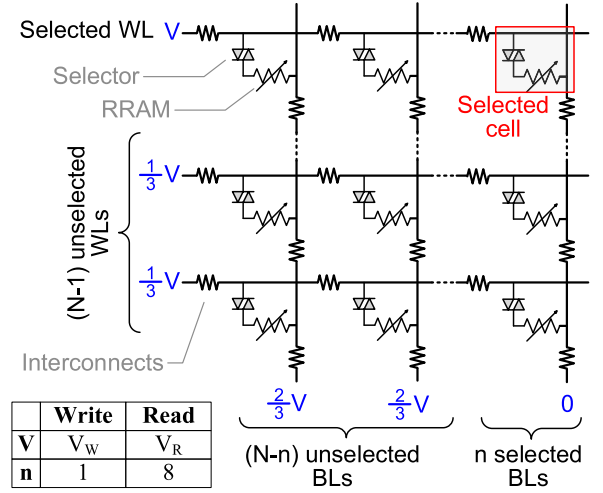


Fig. 9. Bias scheme of the worst case write and read operations in the 1S1R cross-point array architecture. N is the array size on one side, and n is the number of selected BLs. V_W (V_R) is the write (read) voltage.

it is suggested that the $V/3$ scheme normally offers the best read margin [34]. In this $V/3$ scheme, the voltage bias of the selected WL (BL) is $V(0)$, and the voltage bias of the unselected WL (BL) is $V/3$ ($2V/3$). In this way, the access voltage on the selected cell is V , while other unselected cells only see $V/3$. $V = V_W$ (V_R) is the write (read) voltage for the selected cell(s). In this array design, we choose positive V_R , because the SET is more difficult to be triggered, as shown in Fig. 3, and also positive V_R can somewhat compensate the RRAM resistance spontaneous shift toward HRS that is shown in Fig. 4. We also assume only 1 b ($n = 1$) is written, and 1 B ($n = 8$) is readout per operation in this paper.

To evaluate the 1S1R array performance, we consider the worst case data pattern. Taking account the IR drop along the wire interconnects, the selected cell(s) should be located at the corner of the array that is farthest from the WL and BL voltage biases. With the $V/3$ scheme and the excellent selectivity of the FAST selector, we can assume that there is no current flowing through other unselected cells as long as the voltage drop ($V_W/3$) on those cells are below V_{th} . However, for the read operation, V_R needs to be larger than V_{th} , otherwise the selected cells cannot be readout.

From the above discussion, it is shown that the design of V_{th} is closely related to both the write and read operations. For the write operation, a large value of V_{th} is preferred to make all the unselected cells remain OFF, and it should satisfy

$$\frac{1}{3}V_W \leq V_{th} - \Delta V \quad (7)$$

where ΔV is the tolerance for V_{th} variation in selectors. We assume $\Delta V = 0.1$ V in this paper. On the contrary, a small value of V_{th} is preferred for the read operation to ensure that there is enough read access voltage on the worst case cell

$$V_R \geq V_{th} + \Delta V. \quad (8)$$

The tradeoff in designing V_{th} is thus constrained by both the write (SET) operation and the read operation. The SET operation may fail with a small value of V_{th} , while a large value of V_{th} may lead to read disturbance on unselected cells near

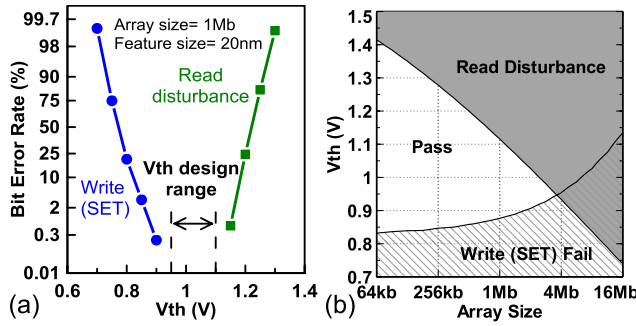


Fig. 10. (a) Bit-error rate with different values of V_{th} of the selector. Since both $V_W/3$ and V_R are determined based on V_{th} , a small value of V_{th} is not sufficient to set the RRAM device, and a large value of V_{th} will easily disturb the resistance states during the read. (b) Pass/fail region with different values of V_{th} and array sizes. Increasing the array size will degrade the array performance as there is more voltage drop on the interconnects. The result suggests that it has the potential to achieve array size ~ 2 Mb.

the WL edge, because the applied voltage at the WL has to be boosted in order to deliver the desired V_R to the selected cells. We run the Monte Carlo simulation using the RRAM compact model with variations [Fig. 5(b)] considered. Fig. 10(a) shows the bit-error rate of the 10-ns write (SET) and read operations with different values of V_{th} . The array size is fixed at 1 Mb (1024×1024), and the feature size of RRAM is 20 nm with the interconnect resistance values from ITRS [28]. By designing the transistors' W/L in the WL driver circuit, the resistance used here can be $R_{LRS} = 100$ k Ω and $R_{HRS} = 1$ M Ω . The bit-error occurrence is defined as either when the SET operation fails (e.g., $R_{LRS} > 200$ k Ω) or V_R is large enough to disturb the resistance state of a cell (e.g., a 5% ΔR change). The results suggest that the favorable V_{th} design point resides in the range from ~ 0.95 to 1.1 V for zero bit-error rate.

Due to the conflicting nature of V_{th} that affects the write and read operations, it is attractive to explore how large the 1S1R cross-point array can support. We simulate the worst case variation corner of the parameters with different array sizes and V_{th} , as shown in Fig. 10(b). As the array size scales up, the V_{th} design range shrinks due to the increased IR drop on longer wire. The result suggests that it has the potential to achieve array size ~ 2 Mb with the IMEC RRAM devices. For array design beyond 2 Mb, increasing the R_{LRS} or reducing the device variation might be the priority task.

V. CONCLUSION

We present a Verilog-A compact model for metal-oxide-based RRAM devices with bipolar switching characteristics. The model simplifies the RRAM switching mechanisms with essential equations. The gap growth/dissolution velocity generally determines the switching characteristics. We fit the model parameters and the variations to the experimental data of the IMEC HfO_x -based RRAM devices. The Verilog-A model is compatible with common SPICE simulators, and we have demonstrated the model in different RRAM array architectures. In the 1T1R configuration, the model is used to find the desirable SET and RESET programming scheme, and it shows that the key challenge for RRAM scaling is the transistor's reliability due to large operation voltages. In the

1S1R cross-point array, the model is used to determine the V_{th} design range and explore the feasibility of a large-scale array in the constraints of the device variations and IR drop along the interconnects. In sum, the model can be a suitable tool to seek design solutions in various RRAM applications, and the Verilog-A codes are available for downloading at [20].

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REFERENCES

- [1] H. Akinaga and H. Shima, "Resistive random access memory (ReRAM) based on metal oxides," *Proc. IEEE*, vol. 98, no. 12, pp. 2237–2251, Dec. 2010.
- [2] H.-S. P. Wong *et al.*, "Metal-oxide RRAM," *Proc. IEEE*, vol. 100, no. 6, pp. 1951–1970, Jun. 2012.
- [3] B. Gao *et al.*, "Unified physical model of bipolar oxide-based resistive switching memory," *IEEE Electron Device Lett.*, vol. 30, no. 12, pp. 1326–1328, Dec. 2009.
- [4] D. Ielmini, "Modeling the universal set/reset characteristics of bipolar RRAM by field- and temperature-driven filament growth," *IEEE Trans. Electron Devices*, vol. 58, no. 12, pp. 4309–4317, Dec. 2011.
- [5] A. Kawahara *et al.*, "An 8 Mb multi-layered cross-point ReRAM macro with 443 MB/s write throughput," *IEEE J. Solid-State Circuits*, vol. 48, no. 1, pp. 178–185, Jan. 2013.
- [6] T.-Y. Liu *et al.*, "A 130.7-mm² 2-layer 32-Gb ReRAM memory device in 24-nm technology," *IEEE J. Solid-State Circuits*, vol. 49, no. 1, pp. 140–153, Jan. 2014.
- [7] G. W. Burr *et al.*, "Access devices for 3D crosspoint memory," *J. Vac. Sci. Technol. B*, vol. 32, no. 4, p. 040802, 2014.
- [8] J. Liang and H.-S. P. Wong, "Cross-point memory array without cell selectors—Device characteristics and data storage pattern dependencies," *IEEE Trans. Electron Devices*, vol. 57, no. 10, pp. 2531–2538, Oct. 2010.
- [9] A. Chen, "A comprehensive crossbar array model with solutions for line resistance and nonlinear device characteristics," *IEEE Trans. Electron Devices*, vol. 60, no. 4, pp. 1318–1326, Apr. 2013.
- [10] S. Long *et al.*, "Compact analytical models for the SET and RESET switching statistics of RRAM inspired in the cell-based percolation model of gate dielectric breakdown," in *Proc. IEEE Int. Rel. Phys. Symp. (IRPS)*, Apr. 2013, pp. 5A.6.1–5A.6.8.
- [11] P. Huang *et al.*, "A physics-based compact model of metal-oxide-based RRAM DC and AC operations," *IEEE Trans. Electron Devices*, vol. 60, no. 12, pp. 4090–4097, Dec. 2013.
- [12] M. Bocquet *et al.*, "Robust compact model for bipolar oxide-based resistive switching memories," *IEEE Trans. Electron Devices*, vol. 61, no. 3, pp. 674–681, Mar. 2014.
- [13] L. Larcher, F. M. Puglisi, P. Pavan, A. Padovani, L. Vandelli, and G. Bersuker, "A compact model of program window in HfO_x RRAM devices for conductive filament characteristics analysis," *IEEE Trans. Electron Devices*, vol. 61, no. 8, pp. 2668–2673, Aug. 2014.
- [14] X. Guan, S. Yu, and H.-S. P. Wong, "A SPICE compact model of metal oxide resistive switching memory with variations," *IEEE Electron Device Lett.*, vol. 33, no. 10, pp. 1405–1407, Oct. 2012.
- [15] Z. Jiang, S. Yu, Y. Wu, J. H. Engel, X. Guan, and H.-S. P. Wong, "Verilog-A compact model for oxide-based resistive random access memory (RRAM)," in *Proc. Int. Conf. Simulation Semiconductor Process. Devices (SISPAD)*, 2014, pp. 41–44.
- [16] W. G. Bennett *et al.*, "Dynamic modeling of radiation-induced state changes in HfO_2/Hf 1T1R RRAM," *IEEE Trans. Nucl. Sci.*, vol. 61, no. 6, pp. 3497–3503, Dec. 2014.
- [17] Y. Y. Chen *et al.*, "Balancing SET/RESET pulse for $> 10^{10}$ endurance in HfO_2/Hf 1T1R bipolar RRAM," *IEEE Trans. Electron Devices*, vol. 59, no. 12, pp. 3243–3249, Dec. 2012.
- [18] Y. Y. Chen *et al.*, "Improvement of data retention in HfO_2/Hf 1T1R RRAM cell under low operating current," in *Proc. IEEE Int. Electron Devices Meeting (IEDM)*, Dec. 2013, pp. 10.1.1–10.1.4.
- [19] A. Fantini *et al.*, "Intrinsic switching variability in HfO_2 RRAM," in *Proc. IEEE 5th Int. Memory Workshop (IMW)*, May 2013, pp. 30–33.
- [20] ASU RRAM Model. [Online]. Available: <http://faculty.engineering.asu.edu/shimengyu/model-downloads/>

- [21] S. Yu, Y. Y. Chen, X. Guan, H.-S. P. Wong, and J. A. Kittl, "A Monte Carlo study of the low resistance state retention of HfO_x based resistive switching memory," *Appl. Phys. Lett.*, vol. 100, no. 4, p. 043507, 2012.
- [22] J. McPherson, J.-Y. Kim, A. Shanware, and H. Mogul, "Thermochemical description of dielectric breakdown in high dielectric constant materials," *Appl. Phys. Lett.*, vol. 82, no. 13, pp. 2121–2123, 2003.
- [23] S. Yu, B. Gao, Z. Fang, H. Yu, J. Kang, and H.-S. P. Wong, "A neuromorphic visual system using RRAM synaptic devices with sub-pJ energy and tolerance to variability: Experimental characterization and large-scale modeling," in *Proc. IEEE Int. Electron Device Meeting (IEDM)*, Dec. 2012, pp. 10.4.1–10.4.4.
- [24] J. C. Butcher, *Numerical Methods for Ordinary Differential Equations*. New York, NY, USA: Wiley, 2005.
- [25] *Predictive Technology Model (PTM)*. [Online]. Available: <http://ptm.asu.edu/>
- [26] E. Yalon, A. A. Sharma, M. Skowronski, J. A. Bain, D. Ritter, and I. V. Karpov, "Thermometry of filamentary RRAM devices," *IEEE Trans. Electron Devices*, vol. 62, no. 9, pp. 2972–2977, Sep. 2015.
- [27] R. Fang, W. Chen, L. Gao, W. Yu, and S. Yu, "Low-temperature characteristics of HfO_x -based resistive random access memory," *IEEE Electron Device Lett.*, vol. 36, no. 6, pp. 567–569, Jun. 2015.
- [28] *International Technology Roadmap for Semiconductors (ITRS)*. [Online]. Available: <http://www.itrs.net/>
- [29] K. Virwani *et al.*, "Sub-30 nm scaling and high-speed operation of fully-confined access-devices for 3D crosspoint memory based on mixed-ionic-electronic-conduction (MIEC) materials," in *Proc. IEEE Int. Electron Device Meeting (IEDM)*, Dec. 2012, pp. 2.7.1–2.7.4.
- [30] S. H. Jo, T. Kumar, S. Narayanan, W. D. Lu, and H. Nazarian, "3D-stackable crossbar resistive memory based on field assisted super-linear threshold (FAST) selector," in *Proc. IEEE Int. Electron Device Meeting (IEDM)*, Dec. 2014, pp. 6.7.1–6.7.4.
- [31] W. Lee *et al.*, "Varistor-type bidirectional switch ($J_{\text{MAX}} > 10^7 \text{ A/cm}^2$, selectivity $\sim 10^4$) for 3D bipolar resistive memory arrays," in *Proc. Symp. VLSI Technol. (VLSIT)*, Jun. 2012, pp. 37–38.
- [32] J.-J. Huang, Y.-M. Tseng, C.-W. Hsu, and T.-H. Hou, "Bipolar non-linear $\text{Ni/TiO}_2/\text{Ni}$ selector for 1S1R crossbar array applications," *IEEE Electron Device Lett.*, vol. 32, no. 10, pp. 1427–1429, Oct. 2011.
- [33] L. Zhang, S. Cosemans, D. J. Wouters, G. Groeseneken, M. Jurczak, and B. Govoreanu, "Selector design considerations and requirements for 1S1R RRAM crossbar array," in *Proc. IEEE 6th Int. Memory Workshop (IMW)*, May 2014, pp. 1–4.
- [34] J. Zhou, K.-H. Kim, and W. Lu, "Crossbar RRAM arrays: Selector device requirements during read operation," *IEEE Trans. Electron Devices*, vol. 61, no. 5, pp. 1369–1376, May 2014.



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